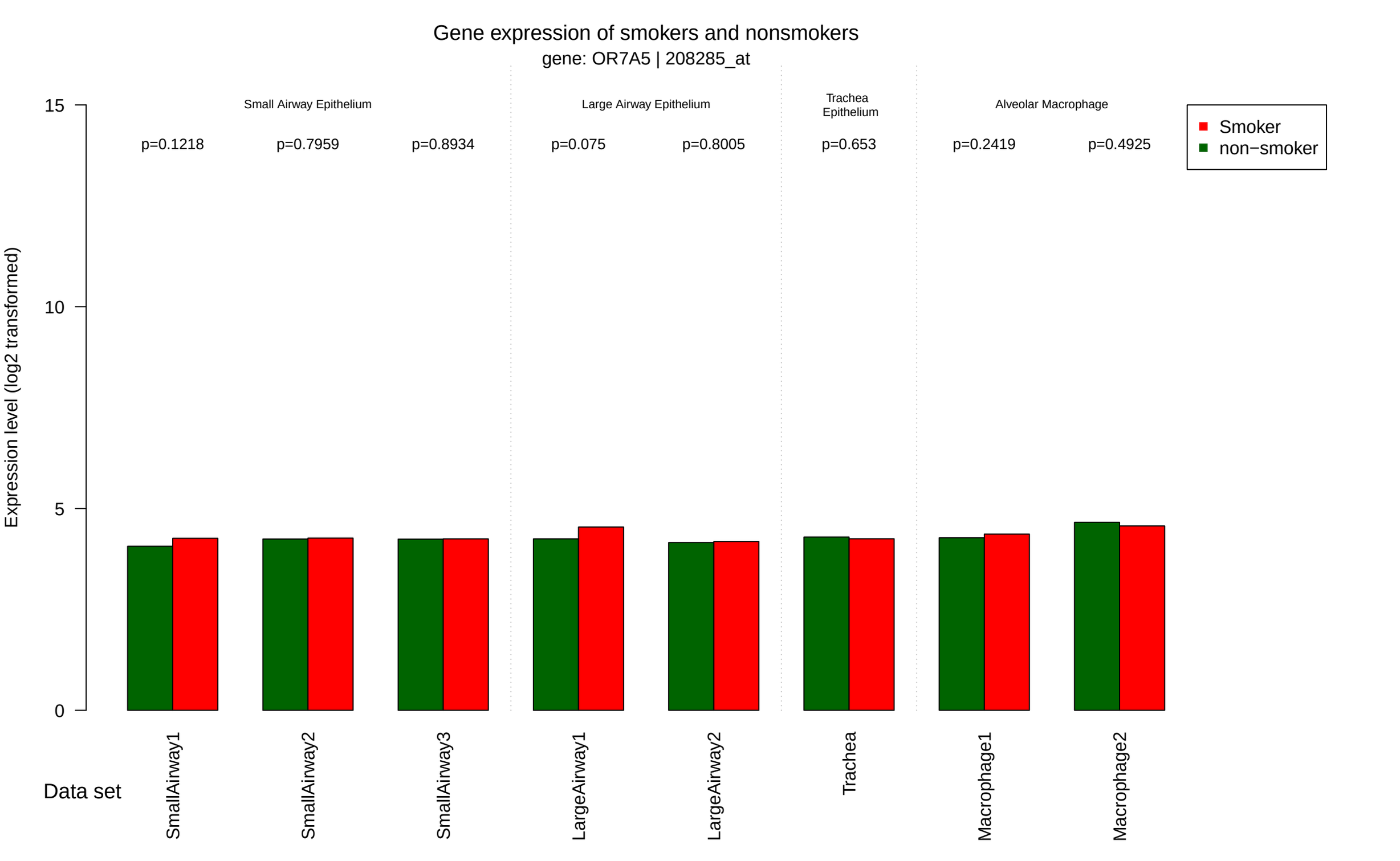
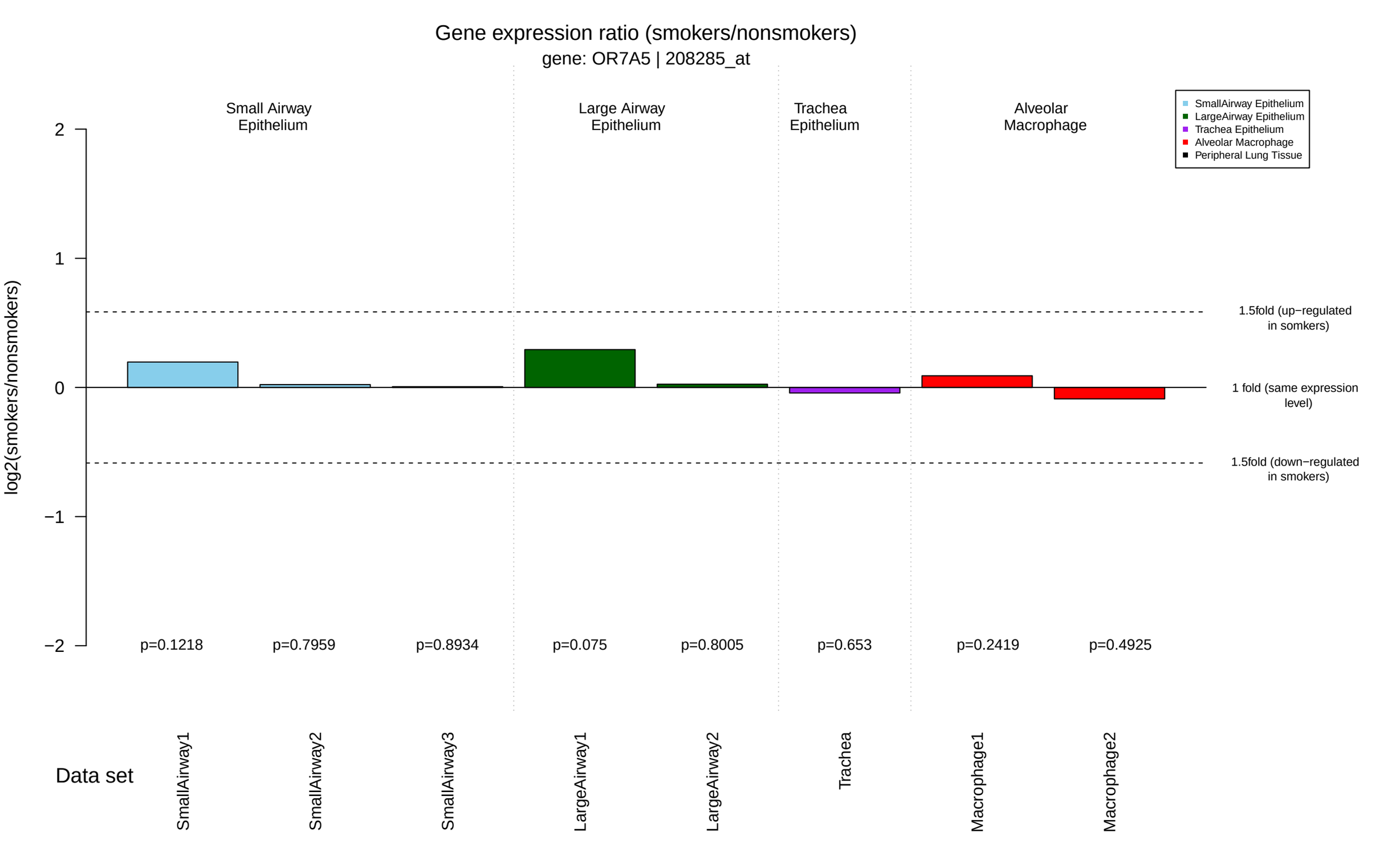
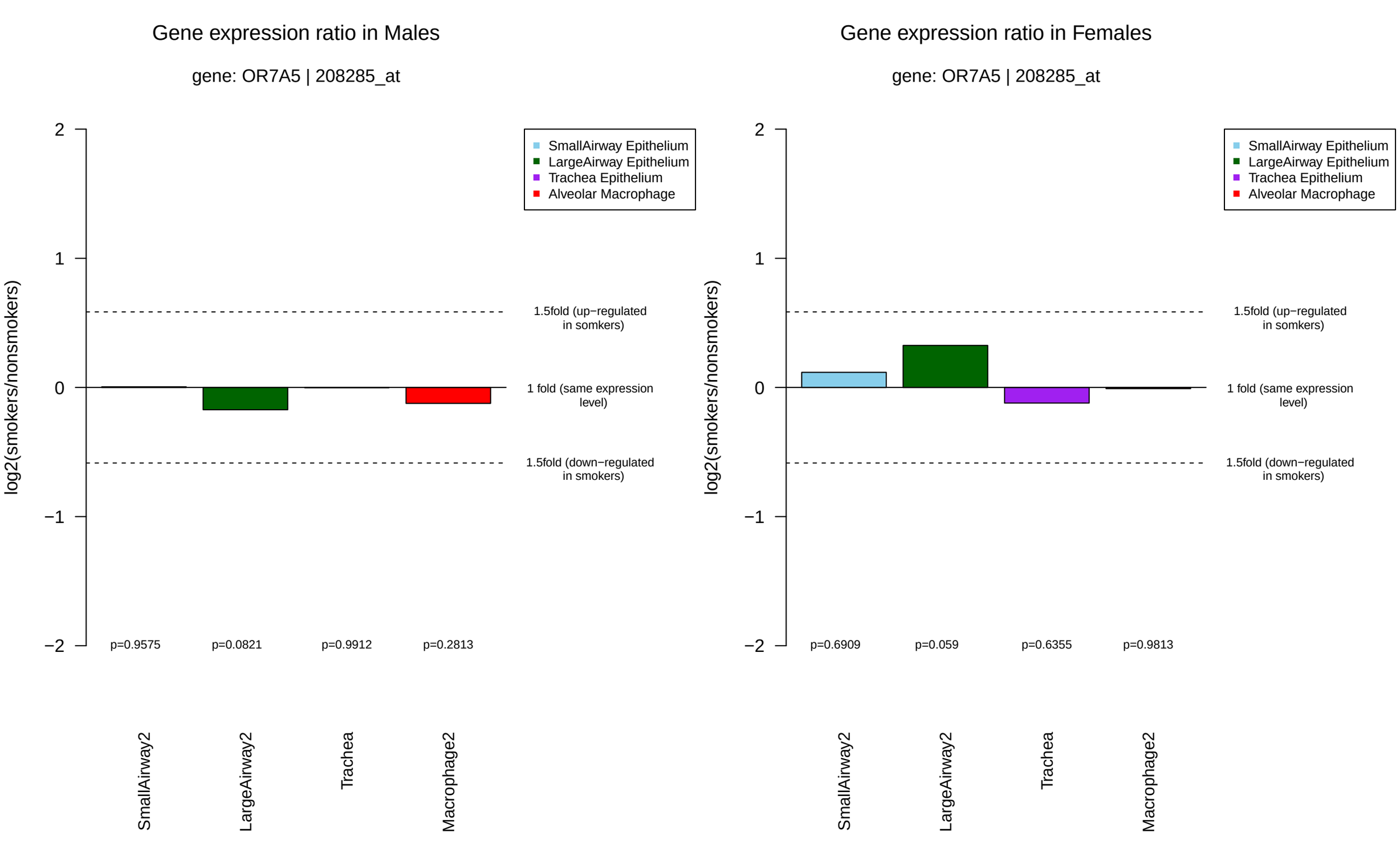
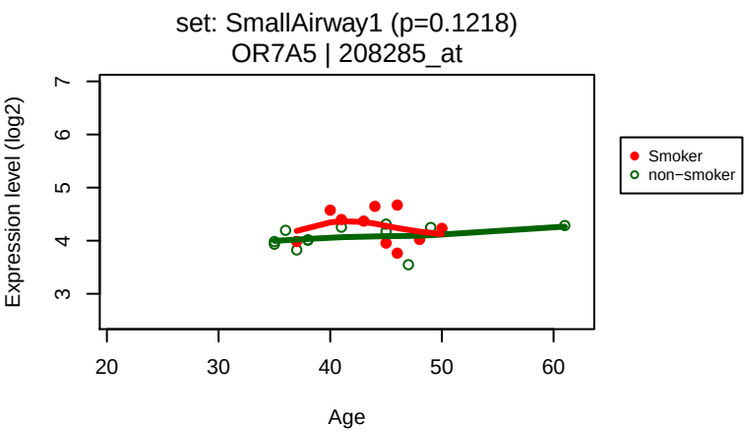
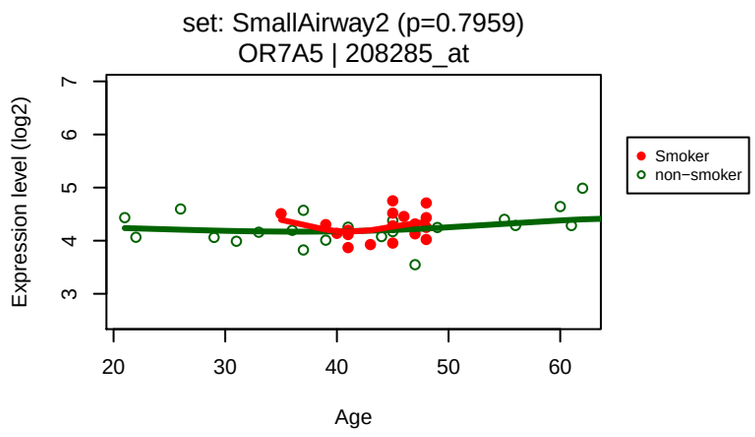
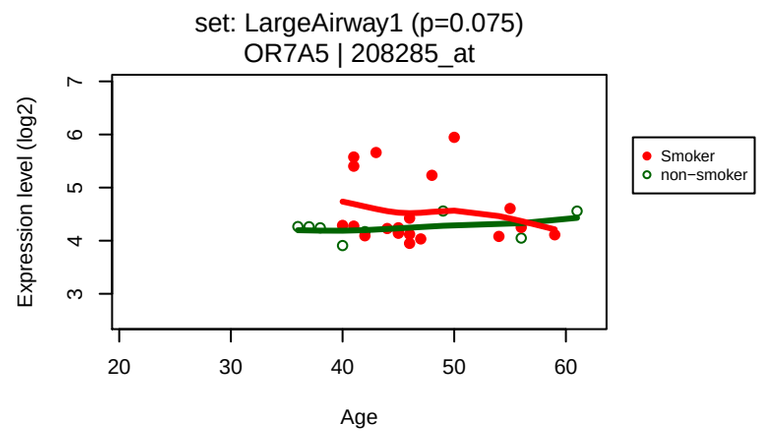

Gene expression of smokers and nonsmokers
gene: OR7A5 | 208285_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1218
p=0.7959
p=0.8934
p=0.075
p=0.8005
p=0.653
p=0.2419
p=0.4925
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

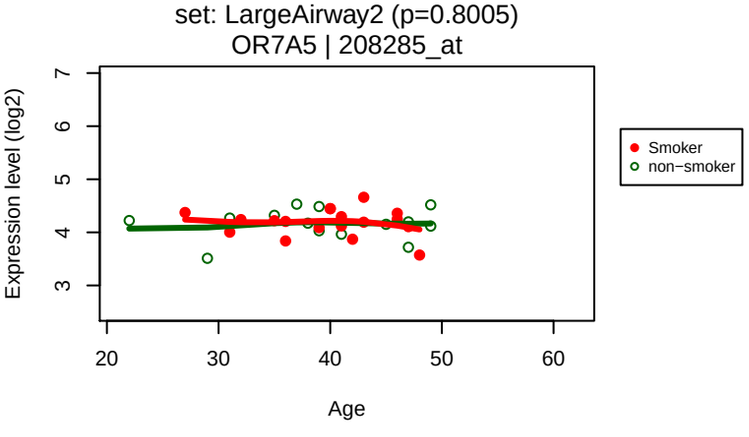

Gene expression ratio (smokers/nonsmokers)
gene: OR7A5 | 208285_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1218
p=0.7959
p=0.8934
p=0.075
p=0.8005
p=0.653
p=0.2419
p=0.4925
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

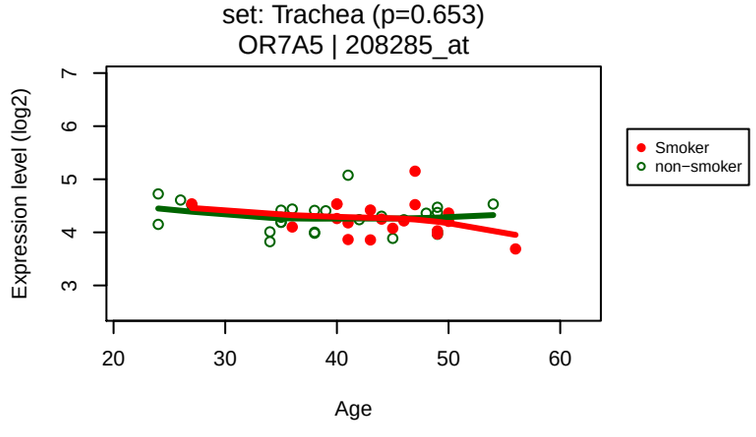

Gene expression ratio in Males
gene: OR7A5 | 208285_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9575
p=0.0821
p=0.9912
p=0.2813
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR7A5 | 208285_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6909
p=0.059
p=0.6355
p=0.9813
SmallAirway2
LargeAirway2
Trachea
Macrophage2

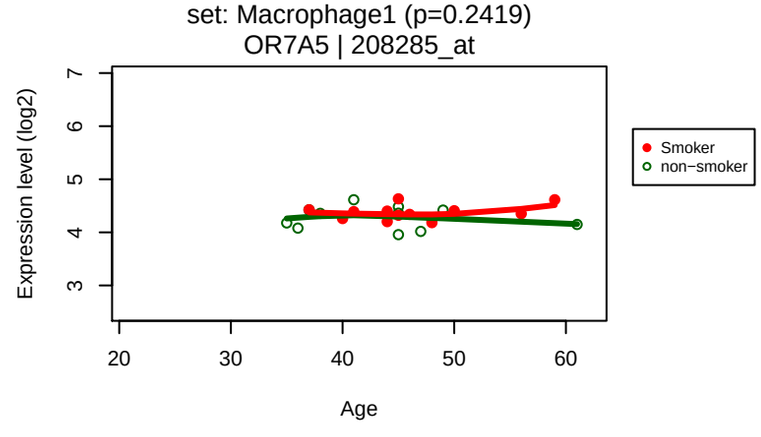

set: SmallAirway1 (p=0.1218)
OR7A5 | 208285_at
Expression level (log2)
Age
Smoker
non−smoker

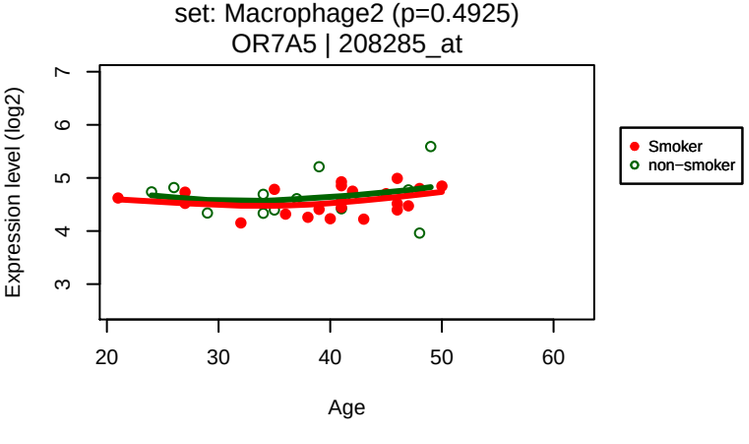

set: SmallAirway2 (p=0.7959)
OR7A5 | 208285_at
Expression level (log2)
Age
Smoker
non−smoker

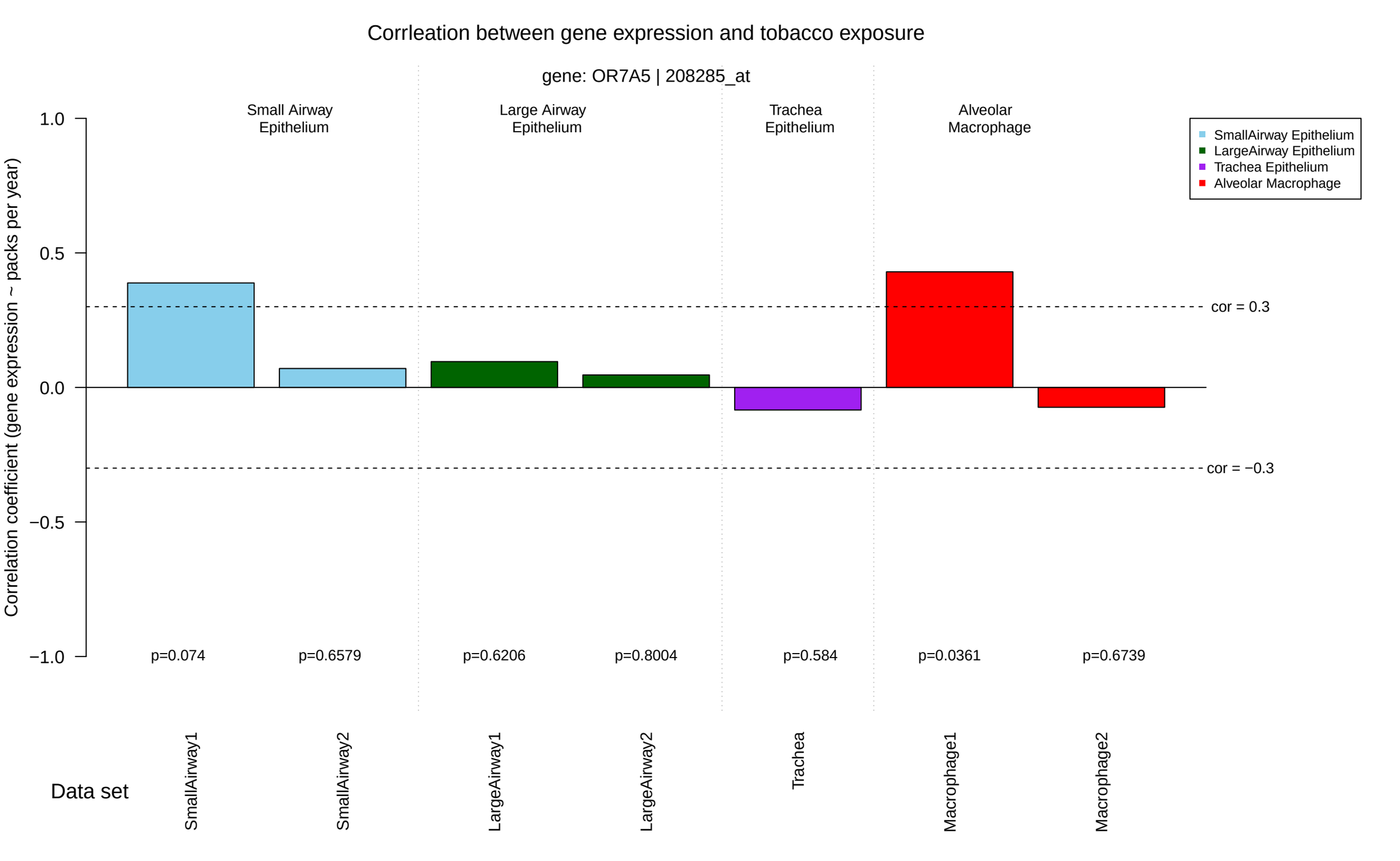

set: LargeAirway1 (p=0.075)
OR7A5 | 208285_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8005)
OR7A5 | 208285_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.653)
OR7A5 | 208285_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2419)
OR7A5 | 208285_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4925)
OR7A5 | 208285_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR7A5 | 208285_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.074
p=0.6579
p=0.6206
p=0.8004
p=0.584
p=0.0361
p=0.6739
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2